

## CHAPTER 13

# SEISMIC DESIGN REQUIREMENTS FOR NONSTRUCTURAL COMPONENTS

### 13.1 GENERAL

**13.1.1 Scope.** This chapter establishes minimum design criteria for nonstructural components that are permanently attached to structures and for their supports and attachments. Where the weight of a nonstructural component is greater than or equal to 25% of the effective seismic weight,  $W$ , of the structure as defined in Section 12.7.2, the component shall be classified as a nonbuilding structure and shall be designed in accordance with Section 15.3.2.

**13.1.2 Seismic Design Category.** For the purposes of this chapter, nonstructural components shall be assigned to the same Seismic Design Category as the structure that they occupy or to which they are attached.

**13.1.3 Component Importance Factor.** All components shall be assigned a component Importance Factor as indicated in this section. The component Importance Factor,  $I_p$ , shall be taken as 1.5 if any of the following conditions apply:

1. The component is required to function for life-safety purposes after an earthquake, including fire protection sprinkler systems and egress stairways.
2. The component conveys, supports, or otherwise contains toxic, highly toxic, or explosive substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released.
3. The component is in or attached to a Risk Category IV structure, and it is needed for continued operation of the facility or its failure could impair the continued operation of the facility.
4. The component conveys, supports, or otherwise contains hazardous substances and is attached to a structure or portion thereof classified by the Authority Having Jurisdiction as a hazardous occupancy.

All other components shall be assigned a component Importance Factor,  $I_p$ , equal to 1.0.

**13.1.4 Exemptions.** The following nonstructural components are exempt from the requirements of this chapter:

1. Furniture except storage cabinets, as noted in Table 13.5-1;
2. Temporary or movable equipment;
3. Architectural components in Seismic Design Category B, other than parapets, provided that the component Importance Factor,  $I_p$ , is equal to 1.0;
4. Mechanical and electrical components in Seismic Design Category B;
5. Mechanical and electrical components in Seismic Design Category C provided that either

- a. The component Importance Factor,  $I_p$ , is equal to 1.0 and the component is positively attached to the structure; or
  - b. The component weighs 20 lb (89 N) or less or, in the case of a distributed system, 5 lb/ft (73 N/m) or less.
6. Discrete mechanical and electrical components in Seismic Design Categories D, E, or F that are positively attached to the structure, provided that either
    - a. The component weighs 400 lb (1,779 N) or less, the center of mass is located 4 ft (1.22 m) or less above the adjacent floor level, flexible connections are provided between the component and associated ductwork, piping, and conduit, and the component Importance Factor,  $I_p$ , is equal to 1.0; or
    - b. The component weighs 20 lb (89 N) or less ~~or, in the case of a distributed system, 5 lb/ft (73 N/m) or less, and~~
  7. Distribution systems in Seismic Design Categories D, E, or F included in the exceptions for conduit, cable tray, and raceways in Section 13.6.5, duct systems in 13.6.6 and piping and tubing systems in 13.6.7.3. Where in-line components, such as valves, in-line suspended pumps, and mixing boxes require independent support, they shall be addressed as discrete components and shall be braced considering the tributary contribution of the attached distribution system.

**13.1.5 Premanufactured Modular Mechanical and Electrical Systems.** Premanufactured mechanical and electrical modules 6 ft (1.8 m) high and taller that are not otherwise prequalified in accordance with Chapter 13 and that contain or support mechanical and electrical components shall be designed in accordance with the provisions for nonbuilding structures similar to buildings in Chapter 15. Nonstructural components contained or supported within modular systems shall be designed in accordance with Chapter 13.

**13.1.6 Application of Nonstructural Component Requirements to Nonbuilding Structures.** Nonbuilding structures (including storage racks and tanks) that are supported by other structures shall be designed in accordance with Chapter 15. Where Section 15.3 requires that seismic forces be determined in accordance with Chapter 13 and values for  $R_p$  are not provided in Table 13.5-1 or 13.6-1,  $R_p$  shall be taken as equal to the value of  $R$  listed in Chapter 15. The value of  $a_p$  shall be determined in accordance with footnote *a* of Table 13.5-1 or 13.6-1.

**13.1.7 Reference Documents.** Where a reference document provides a basis for the earthquake-resistant design of a particular type of nonstructural component, that document is permitted to be used, subject to the approval of the Authority Having Jurisdiction and the following conditions:

1. The design earthquake forces shall not be less than those determined in accordance with Section 13.3.1.
2. Each nonstructural component's seismic interactions with all other connected components and with the supporting structure shall be accounted for in the design. The component shall accommodate drifts, deflections, and relative displacements determined in accordance with the applicable seismic requirements of this standard.
3. Nonstructural component anchorage requirements shall not be less than those specified in Section 13.4.

#### 13.1.8 Reference Documents Using Allowable Stress Design.

Where a reference document provides a basis for the earthquake-resistant design of a particular type of component, and the same reference document defines acceptance criteria in terms of allowable stresses rather than strengths, that reference document is permitted to be used. The allowable stress load combination shall consider dead, live, operating, and earthquake loads in addition to those in the reference document. The earthquake loads determined in accordance with Section 13.3.1 shall be multiplied by a factor of 0.7. The allowable stress design load combinations of Section 2.4 need not be used. The component shall also accommodate the relative displacements specified in Section 13.3.2.

### 13.2 GENERAL DESIGN REQUIREMENTS

#### 13.2.1 Applicable Requirements for Architectural, Mechanical, and Electrical Components, Supports, and Attachments.

Architectural, mechanical, and electrical components, supports, and attachments shall comply with the sections referenced in Table 13.2-1. These requirements shall be satisfied by one of the following methods:

1. Project-specific design and documentation submitted for approval to the Authority Having Jurisdiction after review and acceptance by a registered design professional.
2. Submittal of the manufacturer's certification that the component is seismically qualified by at least one of the following:
  - a. Analysis, or
  - b. Testing in accordance with the alternative set forth in Section 13.2.5, or
  - c. Experience data in accordance with the alternative set forth in Section 13.2.6.

**13.2.2 Special Certification Requirements for Designated Seismic Systems.** Certifications shall be provided for designated seismic systems assigned to Seismic Design Categories C through F as follows:

1. Active mechanical and electrical equipment that must remain operable following the design earthquake ground motion shall be certified by the manufacturer as operable whereby active parts or energized components shall be certified

exclusively on the basis of approved shake table testing in accordance with Section 13.2.5 or experience data in accordance with Section 13.2.6 unless it can be shown that the component is inherently rugged by comparison with similar seismically qualified components. Evidence demonstrating compliance with this requirement shall be submitted for approval to the Authority Having Jurisdiction after review and acceptance by a registered design professional.

2. Components with hazardous substances and assigned a component Importance Factor,  $I_p$ , of 1.5 in accordance with Section 13.1.3 shall be certified by the manufacturer as maintaining containment following the design earthquake ground motion by (1) analysis, (2) approved shake table testing in accordance with Section 13.2.5, or (3) experience data in accordance with Section 13.2.6. Evidence demonstrating compliance with this requirement shall be submitted for approval to the Authority Having Jurisdiction after review and acceptance by a registered design professional.
3. Certification of components through analysis shall be limited to nonactive components and shall be based on seismic demand considering  $R_p/I_p$  equal to 1.0.

**13.2.3 Consequential Damage.** The functional and physical interrelationship of components, their supports, and their effect on each other shall be considered so that the failure of an essential or nonessential architectural, mechanical, or electrical component shall not cause the failure of an essential architectural, mechanical, or electrical component. Where not otherwise established by analysis or test, required clearances for sprinkler system drops and sprigs shall not be less than those specified in Section 13.2.3.1.

**13.2.3.1 Clearances between Equipment, Distribution Systems, Supports, and Sprinkler System Drops and Sprigs.** The installed clearance between any sprinkler drop or sprig and the following items shall be at least 3 in. (75 mm) in all directions:

1. permanently attached equipment including their structural supports and bracing; and
2. other distribution systems, including their structural supports and bracing.

**EXCEPTION:** Sprinklers installed using flexible sprinkler hose need not meet the installed clearance requirement of this section.

**13.2.4 Flexibility.** The design and evaluation of components, their supports, and their attachments shall consider their flexibility and their strength.

**13.2.5 Testing Alternative for Seismic Capacity Determination.** As an alternative to the analytical requirements of Sections 13.2 through 13.6, testing shall be deemed as an

**Table 13.2-1 Applicable Requirements for Architectural, Mechanical, and Electrical Components: Supports and Attachments**

| Nonstructural Element (i.e., Component, Support, Attachment)                       | General Design Requirements<br>(Section 13.2) | Force and Displacement Requirements<br>(Section 13.3) | Attachment Requirements<br>(Section 13.4) | Architectural Component Requirements<br>(Section 13.5) | Mechanical and Electrical Component Requirements<br>(Section 13.6) |
|------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------|-------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|
| Architectural components and supports and attachments for architectural components | X                                             | X                                                     | X                                         | X                                                      |                                                                    |
| Mechanical and electrical components                                               | X                                             | X                                                     | X                                         |                                                        | X                                                                  |
| Supports and attachments for mechanical and electrical components                  | X                                             | X                                                     | X                                         |                                                        | X                                                                  |

acceptable method to determine the seismic capacity of components and their supports and attachments. Seismic qualification by testing based on a nationally recognized testing standard procedure, such as ICC-ES AC 156, acceptable to the Authority Having Jurisdiction shall be deemed to satisfy the design and evaluation requirements provided that the substantiated seismic capacities equal or exceed the seismic demands determined in accordance with Sections 13.3.1 and 13.3.2. For the testing alternative, the maximum seismic demand determined in accordance with Eq. (13.3-2) is not required to exceed  $3.2I_p W_p$ .

**13.2.6 Experience Data Alternative for Seismic Capacity Determination.** As an alternative to the analytical requirements of Sections 13.2 through 13.6, use of experience data shall be deemed as an acceptable method to determine the seismic capacity of components and their supports and attachments. Seismic qualification by experience data based on nationally recognized procedures acceptable to the Authority Having Jurisdiction shall be deemed to satisfy the design and evaluation requirements provided that the substantiated seismic capacities equal or exceed the seismic demands determined in accordance with Sections 13.3.1 and 13.3.2.

**13.2.7 Construction Documents.** Where design of nonstructural components or their supports and attachments is required by Table 13.2-1, such design shall be shown in construction documents prepared by a registered design professional for use by the owner, authorities having jurisdiction, contractors, and inspectors.

### 13.3 SEISMIC DEMANDS ON NONSTRUCTURAL COMPONENTS

#### 13.3.1 Seismic Design Force

**13.3.1.1 Horizontal Force.** The horizontal seismic design force ( $F_p$ ) shall be applied at the component's center of gravity and distributed relative to the component's mass distribution and shall be determined in accordance with Eq. (13.3-1):

$$F_p = \frac{0.4a_p S_{DS} W_p}{\left(\frac{R_p}{I_p}\right)} \left(1 + 2\frac{z}{h}\right) \quad (13.3-1)$$

$F_p$  is not required to be taken as greater than

$$F_p = 1.6S_{DS} I_p W_p \quad (13.3-2)$$

and  $F_p$  shall not be taken as less than

$$F_p = 0.3S_{DS} I_p W_p \quad (13.3-3)$$

where

- $F_p$  = seismic design force;
- $S_{DS}$  = spectral acceleration, short period, as determined from Section 11.4.5;
- $a_p$  = component amplification factor that varies from 1.00 to 2.50 (select appropriate value from Table 13.5-1 or 13.6-1);
- $I_p$  = component Importance Factor that varies from 1.00 to 1.50 (see Section 13.1.3);
- $W_p$  = component operating weight;
- $R_p$  = component response modification factor that varies from 1.00 to 12 (select appropriate value from Table 13.5-1 or 13.6-1);
- $z$  = height in structure of point of attachment of component with respect to the base. For items at or below the base,  $z$  shall be taken as 0. The value of  $z/h$  need not exceed 1.0; and
- $h$  = average roof height of structure with respect to the base.

The overstrength factor,  $\Omega_0$ , in Table 13.5-1 and Table 13.6-1, is applicable only to anchorage of components to concrete and

masonry where required by Section 13.4.2 or the standards referenced therein and shall be applied in accordance with Section 12.4.3. The redundancy factor,  $\rho$ , is permitted to be taken as equal to 1, and the overstrength factors in Table 12.2-1 need not apply.  $F_p$  shall be applied independently in at least two orthogonal horizontal directions in combination with service or operating loads associated with the component, as appropriate. For vertically cantilevered systems, however,  $F_p$  shall be assumed to act in any horizontal direction.

**13.3.1.2 Vertical Force.** The component shall be designed for a concurrent vertical force  $\pm 0.2S_{DS} W_p$ .

**EXCEPTION:** The concurrent vertical seismic force need not be considered for lay-in access floor panels and lay-in ceiling panels.

**13.3.1.3 Nonseismic Loads.** Where nonseismic loads on nonstructural components exceed  $F_p$ , such loads shall govern the strength design, but the detailing requirements and limitations prescribed in this chapter shall apply.

**13.3.1.4 Dynamic Analysis.** In lieu of the forces determined in accordance with Eq. (13.3-1), accelerations used to determine design forces for nonstructural components are permitted to be determined by one of the following dynamic analysis methods:

1. Linear dynamic analysis procedures of Section 12.9,
2. Nonlinear response history procedures of Chapters 16, 17, and 18,
3. Floor response spectra using the procedures in Section 13.3.1.1 or 13.3.1.2. Where the alternate floor response spectra approach of Section 13.3.1.2 is used, the procedures and results shall be subject to an independent peer review in accordance with Section 1.3.1.3.4.

Seismic forces shall be in accordance with Eq. (13.3-4):

$$F_p = \frac{a_i a_p \cancel{R_p} A_x}{\left(\frac{R_p}{I_p}\right)} \quad (13.3-4)$$

Determination of the accelerations of the structure using these methods shall be performed with  $R = 1.0$ . Seismic forces for the linear dynamic analysis or seismic response history procedures shall be calculated in accordance with Eq. (13.3-4), where  $a_i$  is the maximum acceleration at level  $i$  obtained from the modal analysis and where  $A_x$  is the torsional amplification factor determined by Eq. (12.8-14). Where seismic response history analysis with at least seven ground motions are analyzed,  $a_i$  shall be taken as the average of the maximum accelerations. Where fewer than seven motions are used, the maximum acceleration value for each floor shall be based on the maximum value from the ground motions analyzed. Upper and lower limits of  $F_p$  determined by Eqs. (13.3-2) and (13.3-3) shall apply.

**13.3.1.4.1 Floor Response Spectra.** Floor accelerations at any level are permitted to be determined by the calculation of floor response spectra. The floor response spectrum shall be calculated for the design earthquake at each floor level based on a seismic response history analysis in accordance with Section 12.9 or in accordance with the procedures in Chapters 16, 17, or 18. For structures with damping systems designed in accordance with Chapter 18, a response history analysis shall be performed at the design earthquake level using the procedures in Section 18.3. The floor response spectrum shall be calculated for each ground motion record analyzed. The floor acceleration,  $a_i$ , shall be the

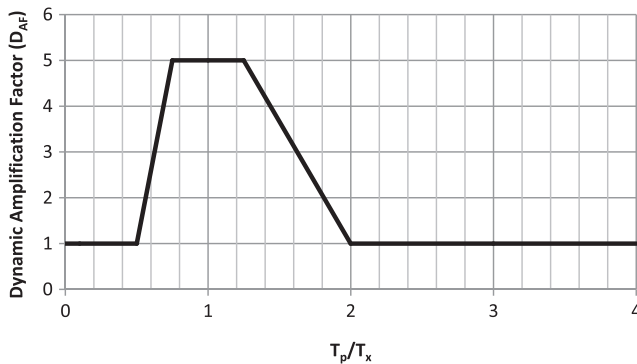


FIGURE 13.3-1 Component Dynamic Amplification Factor

maximum acceleration value from the floor response spectra for the component period, and the value of  $a_p$  shall be taken as 1.0.

**13.3.1.4.2 Alternate Floor Response Spectra.** The periods of vibration and mode shapes of the structure shall be calculated for at least the first three modes in each orthogonal direction using the modal linear dynamic analysis procedure in Section 12.9. The modal participation factors for each of the first three modes shall be calculated in each direction.

The component dynamic amplification factor,  $D_{AF}$ , determined as a ratio of the component period,  $T_p$ , to the building modal period,  $T_x$ , shall be determined based on Fig. 13.3-1.

For each of the first three modes in each direction, the modal acceleration at each floor shall be calculated as a function of the nonstructural component period based on Eq. (13.3-5):

$$A_{ix} = p_{ix} S_{ai} D_{AF} \quad (13.3-5)$$

where  $A_{ix}$  is the floor acceleration for mode  $x$  at level  $i$ ,  $p_{ix}$  is the modal participation factor for mode  $x$  at level  $i$  obtained from the modal analysis,  $S_{ai}$  is the spectral acceleration for mode  $x$ , and  $D_{AF}$  is the dynamic amplification factor as a function of the ratio of component period to building period for mode  $x$  using Fig. 13.3-1.

The floor response spectrum shall be taken as the maximum floor acceleration at each building modal period for at least the first three modes, but not less than the spectral acceleration at the base of the building.

The design seismic horizontal force in each direction for a nonstructural component shall be determined by Eq. (13.3-4) with the product of  $a_i a_p$  replaced by  $A_{ix}$ , the acceleration from the floor response spectrum for the period of vibration of the nonstructural component at level  $i$  on which the nonstructural component is anchored.

**13.3.2 Seismic Relative Displacements.** The effects of seismic relative displacements shall be considered in combination with displacements caused by other loads as appropriate. Seismic relative displacements,  $D_{pl}$ , shall be determined in accordance with Eq. (13.3-6):

$$D_{pl} = D_p I_e \quad (13.3-6)$$

where

$I_e$  = the Importance Factor in Section 11.5.1, and  
 $D_p$  = displacement determined in accordance with the equations set forth in Sections 13.3.2.1 and 13.3.2.2.

**13.3.2.1 Displacements within Structures.** For two connection points on the same structure A or the same structural system,

one at a height  $h_x$  and the other at a height  $h_y$ ,  $D_p$  shall be determined as

$$D_p = \Delta_{xA} - \Delta_{yA} \quad (13.3-7)$$

Alternatively,  $D_p$  is permitted to be determined using linear dynamic procedures described in Section 12.9.  $D_p$  is not required to be taken as greater than

$$D_p = \frac{(h_x - h_y) \Delta_{aA}}{h_{sx}} \quad (13.3-8)$$

**13.3.2.2 Displacements between Structures.** For two connection points on separate structures A and B or separate structural systems, one at a height  $h_x$  and the other at a height  $h_y$ ,  $D_p$  shall be determined as

$$D_p = |\delta_{xA}| + |\delta_{yB}| \quad (13.3-9)$$

$D_p$  is not required to be taken as greater than

$$D_p = \frac{h_x \Delta_{aA}}{h_{sx}} + \frac{h_y \Delta_{aB}}{h_{sx}} \quad (13.3-10)$$

where

$D_p$  = relative seismic displacement that the component must be designed to accommodate;  
 $\delta_{xA}$  = deflection at building level  $x$  of structure A, determined in accordance with Eq. (12.8-15);  
 $\delta_{yA}$  = deflection at building level  $y$  of structure A, determined in accordance with Eq. (12.8-15);  
 $\delta_{yB}$  = deflection at building level  $y$  of structure B, determined in accordance with Eq. (12.8-15);  
 $h_x$  = height of level  $x$  to which upper connection point is attached;  
 $h_y$  = height of level  $y$  to which lower connection point is attached;  
 $\Delta_{aA}$  = allowable story drift for structure A as defined in Table 12.12-1;  
 $\Delta_{aB}$  = allowable story drift for structure B as defined in Table 12.12-1; and  
 $h_{sx}$  = story height used in the definition of the allowable drift  $\Delta_a$  in Table 12.12-1. Note that  $\Delta_a/h_{sx}$  = the drift index.

The effects of seismic relative displacements shall be considered in combination with displacements caused by other loads as appropriate.

**13.3.3 Component Period.** The fundamental period of the nonstructural component (including its supports and attachment to the structure),  $T_p$ , shall be determined by the following equation provided that the component, supports, and attachment can be reasonably represented analytically by a simple spring and mass single degree-of-freedom system:

$$T_p = 2\pi \sqrt{\frac{W_p}{K_{pg}}} \quad (13.3-11)$$

where

$T_p$  = component fundamental period;  
 $W_p$  = component operating weight;



$g$  = gravitational acceleration; and  
 $K_p$  = combined stiffness of the component, supports, and attachments, determined in terms of load per unit deflection at the center of gravity of the component.

Alternatively, the fundamental period of the component,  $T_p$ , in seconds is permitted to be determined from experimental test data or by a properly substantiated analysis.

### 13.4 NONSTRUCTURAL COMPONENT ANCHORAGE

Nonstructural components and their supports shall be attached (or anchored) to the structure in accordance with the requirements of this section, and the attachment shall satisfy the requirements for the parent material as set forth elsewhere in this standard.

Except where permitted in Section 13.6.12, component attachments shall be bolted, welded, or otherwise positively fastened without consideration of frictional resistance produced by the effects of gravity. A continuous load path of sufficient strength and stiffness between the component and the supporting structure shall be provided. Local elements of the structure, including connections, shall be designed and constructed for the component forces where they control the design of the elements or their connections. The component forces shall be those determined in Section 13.3.1. The design documents shall include sufficient information relating to the attachments to verify compliance with the requirements of this section.

**13.4.1 Design Force in the Attachment.** The force in the attachment shall be determined based on the prescribed forces and displacements for the component as determined in Sections 13.3.1 and 13.3.2, except that  $R_p$  shall not be taken as larger than 6.

#### 13.4.2 Anchors in Concrete or Masonry

**13.4.2.1 Anchors in Concrete.** Anchors in concrete shall be designed in accordance with Chapter 17 of ACI 318.

**13.4.2.2 Anchors in Masonry.** Anchors in masonry shall be designed in accordance with TMS 402. Anchors shall be designed to be governed by the tensile or shear strength of a ductile steel element.

**EXCEPTION:** Anchors shall be permitted to be designed so that either

1. the support or component that the anchor is connecting to the structure undergoes ductile yielding at a load level corresponding to anchor forces not greater than the design strength of the anchors, or
2. the anchors shall be designed to resist the load combinations in accordance with Section 12.4.3 including  $\Omega_0$  as given in Tables 13.5-1 and 13.6-1.

**13.4.2.3 Post-Installed Anchors in Concrete and Masonry.** Post-installed anchors in concrete shall be prequalified for seismic applications in accordance with ACI 355.2 or other approved qualification procedures. Post-installed anchors in masonry shall be prequalified for seismic applications in accordance with approved qualification procedures.

**13.4.3 Installation Conditions.** Determination of forces in attachments shall take into account the expected conditions of installation, including eccentricities and prying effects.

**13.4.4 Multiple Attachments.** Determination of force distribution of multiple attachments at one location shall take into account the stiffness and ductility of the component,

component supports, attachments, and structure and the ability to redistribute loads to other attachments in the group. Designs of anchorage in concrete in accordance with Chapter 17 of ACI 318 shall be considered to satisfy this requirement.

**13.4.5 Power-Actuated Fasteners.** Power-actuated fasteners in concrete or steel shall not be used for sustained tension loads or for brace applications in Seismic Design Categories D, E, or F unless approved for seismic loading. Power-actuated fasteners in masonry are not permitted unless approved for seismic loading.

#### EXCEPTIONS:

1. Power-actuated fasteners in concrete used for support of acoustical tile or lay-in panel suspended ceiling applications and distributed systems where the service load on any individual fastener does not exceed 90 lb (400 N), and
2. Power-actuated fasteners in steel where the service load on any individual fastener does not exceed 250 lb (1,112 N).

**13.4.6 Friction Clips.** Friction clips in Seismic Design Categories D, E, or F shall not be used for supporting permanent loads in addition to resisting seismic forces. C-type beam and large flange clamps are permitted for hangers provided that they are equipped with restraining straps equivalent to those specified in NFPA 13, Section 9.3.7. Lock nuts or equivalent shall be provided to prevent loosening of threaded connections.

### 13.5 ARCHITECTURAL COMPONENTS

**13.5.1 General.** Architectural components, and their supports and attachments, shall satisfy the requirements of this section. Appropriate coefficients shall be selected from Table 13.5-1.

**EXCEPTION:** Components supported by chains or otherwise suspended from the structure are not required to satisfy the seismic force and relative displacement requirements provided that they meet all of the following criteria:

1. The design load for such items shall be equal to 1.4 times the operating weight acting down with a simultaneous horizontal load equal to 1.4 times the operating weight. The horizontal load shall be applied in the direction that results in the most critical loading for design.
2. Seismic interaction effects shall be considered in accordance with Section 13.2.3.
3. The connection to the structure shall allow a 360-deg range of motion in the horizontal plane.

**13.5.2 Forces and Displacements.** All architectural components, and their supports and attachments, shall be designed for the seismic forces defined in Section 13.3.1.

Architectural components that could pose a life-safety hazard shall be designed to accommodate the seismic relative displacement requirements of Section 13.3.2. Architectural components shall be designed considering vertical deflection caused by joint rotation of cantilever structural members.

**13.5.3 Exterior Nonstructural Wall Elements and Connections.** Exterior nonstructural wall panels or elements that are attached to or enclose the structure shall be designed to accommodate the seismic relative displacements defined in Section 13.3.2 and movements caused by temperature changes.

**Table 13.5-1 Coefficients for Architectural Components**

| Architectural Component                                                                                                     | $a_p^a$        | $R_p$ | $\Omega_0^b$ |
|-----------------------------------------------------------------------------------------------------------------------------|----------------|-------|--------------|
| Interior nonstructural walls and partitions <sup>c</sup>                                                                    |                |       |              |
| Plain (unreinforced) masonry walls                                                                                          | 1              | 1½    | 1½           |
| All other walls and partitions                                                                                              | 1              | 2½    | 2            |
| Cantilever elements (unbraced or braced to structural frame below its center of mass)                                       |                |       |              |
| Parapets and cantilever interior nonstructural walls                                                                        | 2½             | 2½    | 2            |
| Chimneys where laterally braced or supported by the structural frame                                                        | 2½             | 2½    | 2            |
| Cantilever elements (braced to structural frame above its center of mass)                                                   |                |       |              |
| Parapets                                                                                                                    | 1              | 2½    | 2            |
| Chimneys                                                                                                                    | 1              | 2½    | 2            |
| Exterior nonstructural walls <sup>c</sup>                                                                                   | 1 <sup>b</sup> | 2½    | 2            |
| Exterior nonstructural wall elements and connections <sup>b</sup>                                                           |                |       |              |
| Wall element                                                                                                                | 1              | 2½    | NA           |
| Body of wall panel connections                                                                                              | 1              | 2½    | NA           |
| Fasteners of the connecting system                                                                                          | 1¼             | 1     | 1            |
| Veneer                                                                                                                      |                |       |              |
| Limited deformability elements and attachments                                                                              | 1              | 2½    | 2            |
| Low-deformability elements and attachments                                                                                  | 1              | 1½    | 2            |
| Penthouses (except where framed by an extension of the building frame)                                                      | 2½             | 3½    | 2            |
| Ceilings                                                                                                                    |                |       |              |
| All                                                                                                                         | 1              | 2½    | 2            |
| Cabinets                                                                                                                    |                |       |              |
| Permanent floor-supported storage cabinets more than 6 ft (1,829 mm) tall, including contents                               | 1              | 2½    | 2            |
| Permanent floor-supported library shelving, book stacks, and bookshelves more than 6 ft (1,829 mm) tall, including contents | 1              | 2½    | 2            |
| Laboratory equipment                                                                                                        | 1              | 2½    | 2            |
| Access floors                                                                                                               |                |       |              |
| Special access floors (designed in accordance with Section 13.5.7.2)                                                        | 1              | 2½    | 2            |
| All other                                                                                                                   | 1              | 1½    | 1½           |
| Appendages and ornamentations                                                                                               | 2½             | 2½    | 2            |
| Signs and Billboards                                                                                                        | 2½             | 3     | 2            |
| Other rigid components                                                                                                      |                |       |              |
| High-deformability elements and attachments                                                                                 | 1              | 3½    | 2            |
| Limited-deformability elements and attachments                                                                              | 1              | 2½    | 2            |
| Low-deformability materials and attachments                                                                                 | 1              | 1½    | 1½           |
| Other flexible components                                                                                                   |                |       |              |
| High-deformability elements and attachments                                                                                 | 2½             | 3½    | 2½           |
| Limited-deformability elements and attachments                                                                              | 2½             | 2½    | 2½           |
| Low-deformability materials and attachments                                                                                 | 2½             | 1½    | 1½           |
| Egress stairways not part of the building seismic force-resisting system                                                    | 1              | 2½    | 2            |
| Egress stairs and ramp fasteners and attachments                                                                            | 2½             | 2½    | 2½           |

<sup>a</sup>A lower value for  $a_p$  shall not be used unless justified by detailed dynamic analysis. The value for  $a_p$  shall not be less than 1. The value of  $a_p = 1$  is for rigid components and rigidly attached components. The value of  $a_p = 2½$  is for flexible components and flexibly attached components.

<sup>b</sup>Overstrength where required for nonductile anchorage to concrete and masonry. See Section 12.4.3 for seismic load effects including overstrength.

<sup>c</sup>Where flexible diaphragms provide lateral support for concrete or masonry walls and partitions, the design forces for anchorage to the diaphragm shall be as specified in Section 12.11.2.

Such elements shall be supported by means of positive and direct structural supports or by mechanical connections and fasteners in accordance with the following requirements:

1. Connections and panel joints shall allow for the story drift caused by relative seismic displacements ( $D_{pl}$ )

determined in Section 13.3.2, or 0.5 in. (13 mm), whichever is greater.

2. Connections accommodating story drift through sliding mechanisms or bending of threaded steel rods shall satisfy the following:
  - a. Threaded rods or bolts shall be fabricated of low-carbon or stainless steel. Where cold-worked carbon steel threaded rods are used, the rods as fabricated shall meet or exceed the reduction of area, elongation, and tensile strength requirements of ASTM F1554, Grade 36. Grade 55 rods shall also be permitted provided that they meet the requirements of Supplement 1; and
  - b. Where threaded rods connecting the panel to the supports are used in connections using slotted or oversize holes, the rods shall have length to diameter ratios of 4 or less, where the length is the clear distance between the nuts or threaded plates. The slots or oversized holes shall be proportioned to accommodate the full in-plane design story drift in each direction, the nuts shall be installed finger-tight, and a positive means to prevent the nut from backing off shall be used; and
  - c. Connections that accommodate story drift by bending of threaded rods shall satisfy Eq. (13.5-1):

$$(L/d)/D_{pl} \geq 6.0[1/\text{in.}] \quad (13.5-1)$$

$$(L/d)/D_{pl} \geq 0.24[1/\text{mm}] \quad (13.5-1.\text{si})$$

where:

$L$  = clear length of rod between nuts or threaded plates [in. (mm)];

$d$  = rod diameter [in. (mm)]; and

$D_{pl}$  = relative seismic displacement that the connection must be designed to accommodate [in. (mm)].

3. The connecting member itself shall have sufficient ductility and rotation capacity to preclude fracture of the concrete or brittle failures at or near welds.
4. All fasteners in the connecting system such as bolts, inserts, welds, and dowels, and the body of the connectors shall be designed for the force ( $F_p$ ) determined by Section 13.3.1 with values of  $R_p$ ,  $a_p$ , and  $\Omega_0$  taken from Table 13.5-1 applied at the center of mass of the panel. The connecting system shall include both the connections between the wall panels or elements and the structure and the interconnections between wall panels or elements.
5. Where anchorage is achieved using flat straps embedded in concrete or masonry, such straps shall be attached to or hooked around reinforcing steel or otherwise terminated so as to effectively transfer forces to the reinforcing steel or to ensure that pullout of anchorage is not the initial failure mechanism.

**13.5.4 Glass.** Glass in glazed curtain walls and storefronts shall be designed and installed to accommodate without breakage or dislodgement the relative displacement requirement of Section 13.5.9.

Where glass is secured to the window system framing by means of structural sealant glazing, the requirements contained in the reference standards listed in Table 13.5-2 shall also apply.

**13.5.5 Out-of-Plane Bending.** Transverse or out-of-plane bending or deformation of a component or system that is subjected to forces as determined in Section 13.5.2 shall

**Table 13.5-2 Reference Standards for Structural Sealant Glazing**

| ASTM                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------|
| C1087-00 <i>Test Method for Determining Compatibility of Liquid-Applied Sealants with Accessories Used in Structural Glazing Systems</i> |
| C1135-00 <i>Test Method for Determining Tensile Adhesion Properties of Structural Sealants</i>                                           |
| C1184-14 <i>Specification for Structural Silicone Sealants</i>                                                                           |
| C1265-94 <i>Test Method for Determining the Tensile Properties of an Insulating Glass Edge Seal for Structural Glazing Applications</i>  |
| C1294-07 <i>Test Method for Compatibility of Insulating Glass Edge Sealants with Liquid-Applied Glazing Materials</i>                    |
| C1369-07 <i>Specification for Secondary Edge Sealants for Structurally Glazed Insulating Glass Units</i>                                 |

not exceed the deflection capability of the component or system.

**13.5.6 Suspended Ceilings.** Suspended ceilings shall be in accordance with this section.

#### EXCEPTIONS:

1. Suspended ceilings with areas less than or equal to 144 ft<sup>2</sup> (13.4 m<sup>2</sup>) that are surrounded by walls or soffits that are laterally braced to the structure above are exempt from the requirements of this section.
2. Suspended ceilings constructed of screw- or nail-attached gypsum board on one level that are surrounded by and connected to walls or soffits that are laterally braced to the structure above are exempt from the requirements of this section.

**13.5.6.1 Seismic Forces.** The weight of the ceiling,  $W_p$ , shall include the ceiling grid; ceiling tiles or panels; light fixtures if attached to, clipped to, or laterally supported by the ceiling grid; and other components that are laterally supported by the ceiling.  $W_p$  shall be taken as not less than 4 psf (192 N/m<sup>2</sup>).

The seismic force,  $F_p$ , shall be transmitted through the ceiling attachments to the building structural elements or the ceiling–structure boundary.

**13.5.6.2 Industry Standard Construction for Acoustical Tile or Lay-In Panel Ceilings.** Unless designed in accordance with Section 13.5.6.3, or seismically qualified in accordance with Section 13.2.5 or 13.2.6, acoustical tile or lay-in panel ceilings shall be designed and constructed in accordance with this section.

**13.5.6.2.1 Seismic Design Category C.** Acoustical tile or lay-in panel ceilings in structures assigned to Seismic Design Category C shall be designed and installed in accordance with ASTM C635, ASTM C636, and ASTM E580, Section 4—Seismic Design Category C.

**13.5.6.2.2 Seismic Design Categories D through F.** Acoustical tile or lay-in panel ceilings in structures assigned to Seismic Design Categories D, E, and F shall be designed and installed in accordance with ASTM C635, ASTM C636, and ASTM E580, Section 5—Seismic Design Categories D, E, and F as modified by this section.

Acoustical tile or lay-in panel ceilings shall also comply with the following:

- a. The width of the perimeter supporting closure angle or channel shall be not less than 2.0 in. (50 mm) unless qualified perimeter supporting clips are used. Closure angles

or channels shall be screwed or otherwise positively attached to wall studs or other supporting structures. Perimeter supporting clips shall be qualified in accordance with approved test criteria per Section 13.2.5. Perimeter supporting clips shall be attached to the supporting closure angle or channel with a minimum of two screws per clip and shall be installed around the entire ceiling perimeter. In each orthogonal horizontal direction, one end of the ceiling grid shall be attached to the closure angle, channel, or perimeter supporting clip. The other end of the ceiling grid in each horizontal direction shall have a minimum 0.75-in. (19-mm) clearance from the wall and shall rest upon and be free to slide on a closure angle, channel, or perimeter supporting clip.

- b. For ceiling areas exceeding 2,500 ft<sup>2</sup> (232 m<sup>2</sup>), a seismic separation joint or full-height partition that breaks the ceiling up into areas not exceeding 2,500 ft<sup>2</sup> (232 m<sup>2</sup>), each with a ratio of the long to short dimension less than or equal to 4, shall be provided unless structural analyses are performed of the ceiling bracing system for the prescribed seismic forces which demonstrate that ceiling penetrations and closure angles or channels provide sufficient clearance to accommodate the anticipated lateral displacement. Each area shall be provided with closure angles or channels in accordance with Section 13.5.6.2.2.a and horizontal restraints or bracing.

**13.5.6.3 Integral Construction.** As an alternate to providing large clearances around sprinkler system penetrations through ceilings, the sprinkler system and ceiling grid are permitted to be designed and tied together as an integral unit. Such a design shall consider the mass and flexibility of all elements involved, including the ceiling, sprinkler system, light fixtures, and mechanical (HVACR) appurtenances. Such design shall be performed by a registered design professional.

#### 13.5.7 Access Floors

**13.5.7.1 General.** The weight of the access floor,  $W_p$ , shall include the weight of the floor system, 100% of the weight of all equipment fastened to the floor, and 25% of the weight of all equipment supported by but not fastened to the floor. The seismic force,  $F_p$ , shall be transmitted from the top surface of the access floor to the supporting structure.

Overtuning effects of equipment fastened to the access floor panels also shall be considered. The ability of “slip on” heads for pedestals shall be evaluated for suitability to transfer overturning effects of equipment.

Where checking individual pedestals for overturning effects, the maximum concurrent axial load shall not exceed the portion of  $W_p$  assigned to the pedestal under consideration.

**13.5.7.2 Special Access Floors.** Access floors shall be considered to be “special access floors” if they are designed to comply with the following considerations:

1. Connections transmitting seismic loads consist of mechanical fasteners, anchors satisfying the requirements of Chapter 17 of ACI 318, welding, or bearing. Design load capacities comply with recognized design codes and/or certified test results.
2. Seismic loads are not transmitted by friction, power-actuated fasteners, adhesives, or by friction produced solely by the effects of gravity.
3. The design analysis of the bracing system includes the destabilizing effects of individual members buckling in compression.

4. Bracing and pedestals are of structural or mechanical shapes produced to ASTM specifications that specify minimum mechanical properties. Electrical tubing shall not be used.
5. Floor stringers that are designed to carry axial seismic loads and that are mechanically fastened to the supporting pedestals are used.

### 13.5.8 Partitions

**13.5.8.1 General.** Partitions that are tied to the ceiling and all partitions greater than 6 ft (1.8 m) high shall be laterally braced to the building structure. Such bracing shall be independent of any ceiling lateral force bracing. Bracing shall be spaced to limit horizontal deflection at the partition head to be compatible with ceiling deflection requirements as determined in Section 13.5.6 for suspended ceilings and elsewhere in this section for other systems.

**EXCEPTION:** Partitions that meet all of the following conditions:

1. The partition height does not exceed 9 ft (2,740 mm).
2. The linear weight of the partition does not exceed the product of 10 lb (0.479 kN) times the height [ft (m)] of the partition.
3. The partition horizontal seismic load does not exceed 5 psf (0.24 kN/m<sup>2</sup>).

**13.5.8.2 Glass.** Glass in glazed partitions shall be designed and installed in accordance with Section 13.5.9.

### 13.5.9 Glass in Glazed Curtain Walls, Glazed Storefronts, and Glazed Partitions

**13.5.9.1 General.** Glass in glazed curtain walls, glazed storefronts, and glazed partitions shall meet the relative displacement requirement of Eq. (13.5-2):

$$\Delta_{\text{fallout}} \geq 1.25D_{pl} \quad (13.5-2)$$

or 0.5 in. (13 mm), whichever is greater, where:

$\Delta_{\text{fallout}}$  = the relative seismic displacement (drift) at which glass fallout from the curtain wall, storefront wall, or partition occurs (Section 13.5.9.2);

$D_{pl}$  = the relative seismic displacement that the component must be designed to accommodate (Section 13.3.2) ( $D_{pl}$  shall be applied over the height of the glass component under consideration); and

$I_e$  = the Importance Factor determined in accordance with Section 11.5.1.

#### EXCEPTIONS:

1. Glass with sufficient clearances from its frame such that physical contact between the glass and frame does not occur at the design drift, as demonstrated by Eq. (13.5-3), need not comply with this requirement:

$$D_{\text{clear}} \geq 1.25D_{pl} \quad (13.5-3)$$

where  $D_{\text{clear}}$  = relative horizontal (drift) displacement, measured over the height of the glass panel under consideration, which causes initial glass-to-frame contact. For rectangular glass panels within a rectangular wall frame,

$$D_{\text{clear}} = 2c_1 \left( 1 + \frac{h_p c_2}{b_p c_1} \right)$$

where

$h_p$  = the height of the rectangular glass panel;

$b_p$  = the width of the rectangular glass panel;

$c_1$  = the average of the clearances (gaps) on both sides between the vertical glass edges and the frame; and

$c_2$  = the average of the clearances (gaps) at the top and bottom between the horizontal glass edges and the frame.

2. Fully tempered monolithic glass in Risk Categories I, II, and III located no more than 10 ft (3 m) above a walking surface need not comply with this requirement.
3. Annealed or heat-strengthened laminated glass in single thickness with interlayer no less than 0.030 in. (0.76 mm) that is captured mechanically in a wall system glazing pocket, and whose perimeter is secured to the frame by a wet-glazed, gunable, curing elastomeric sealant perimeter bead of 0.5 in. (13 mm) minimum glass contact width, or other approved anchorage system need not comply with this requirement.

**13.5.9.2 Seismic Drift Limits for Glass Components.**  $\Delta_{\text{fallout}}$ , the drift causing glass fallout from the curtain wall, storefront, or partition, shall be determined in accordance with AAMA 501.6 or by engineering analysis.

**13.5.10 Egress Stairs and Ramps.** Egress stairs and ramps not part of the seismic force-resisting system of the structure to which they are attached shall be detailed to accommodate the seismic relative displacements,  $D_{pl}$ , defined in Section 13.3.2 including diaphragm deformation. The net relative displacement shall be assumed to occur in any horizontal direction. Such elements shall be supported by means of positive and direct structural supports or by mechanical connections and fasteners in accordance with the following requirements:

- a. Sliding connections with slotted or oversize holes, sliding bearing supports with keeper assemblies or end stops, and connections that permit movement by deformation of metal attachments, shall accommodate a displacement  $D_{pl}$ , but not less than 0.5 in. (13 mm), without loss of vertical support or inducement of displacement-related compression forces in the stair.
- b. Sliding bearing supports without keeper assemblies or end stops shall be designed to accommodate a displacement  $1.5D_{pl}$ , but not less than 1.0 in. (25 mm) without loss of vertical support. Breakaway restraints are permitted if their failure does not lead to loss of vertical support.
- c. Metal supports shall be designed with rotation capacity to accommodate seismic relative displacements as defined in item b. The strength of such metal supports shall not be limited by bolt shear, weld fracture, or other brittle modes.
- d. All fasteners and attachments such as bolts, inserts, welds, dowels, and anchors shall be designed for the seismic design forces determined in accordance with Section 13.3.1 with  $R_p$ ,  $a_p$ , and  $\Omega_0$  as given in Table 13.5-1.

**EXCEPTION:** If sliding or ductile connections are not provided to accommodate seismic relative displacements, the stiffness and strength of the stair or ramp structure shall be included in the building structural model of Section 12.7.3, and



the stair shall be designed with  $\Omega_0$  corresponding to the seismic force-resisting system but not less than  $2 - 1/2$ .

### 13.6 MECHANICAL AND ELECTRICAL COMPONENTS

**13.6.1 General.** Mechanical and electrical components and their supports shall satisfy the requirements of this section. The attachment of mechanical and electrical components and their supports to the structure shall meet the requirements of Section 13.4. Appropriate coefficients shall be selected from Table 13.6-1.

**EXCEPTION:** Light fixtures, lighted signs, and ceiling fans not connected to ducts or piping, which are supported by chains or otherwise suspended from the structure, are not required to satisfy the seismic force and relative displacement requirements provided that they meet all of the following criteria:

1. The design load for such items shall be equal to 1.4 times the operating weight acting down with a simultaneous horizontal load equal to 1.4 times the operating weight. The horizontal load shall be applied in the direction that results in the most critical loading for the design.
2. Seismic interaction effects shall be considered in accordance with Section 13.2.3.

**Table 13.6-1 Seismic Coefficients for Mechanical and Electrical Components**

| Components                                                                                                                                                                                                                                            | $a_p^a$ | $R_p^b$ | $\Omega_0^c$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|--------------|
| <b>MECHANICAL AND ELECTRICAL COMPONENTS</b>                                                                                                                                                                                                           |         |         |              |
| Air-side HVACR, fans, air handlers, air conditioning units, cabinet heaters, air distribution boxes, and other mechanical components constructed of sheet metal framing                                                                               | 2½      | 6       | 2            |
| Wet-side HVACR, boilers, furnaces, atmospheric tanks and bins, chillers, water heaters, heat exchangers, evaporators, air separators, manufacturing or process equipment, and other mechanical components constructed of high-deformability materials | 1       | 2½      | 2            |
| Air coolers (fin fans), air-cooled heat exchangers, condensing units, dry coolers, remote radiators and other mechanical components elevated on integral structural steel or sheet metal supports                                                     | 2½      | 3       | 1½           |
| Engines, turbines, pumps, compressors, and pressure vessels not supported on skirts and not within the scope of Chapter 15                                                                                                                            | 1       | 2½      | 2            |
| Skirt-supported pressure vessels not within the scope of Chapter 15                                                                                                                                                                                   | 2½      | 2½      | 2            |
| Elevator and escalator components                                                                                                                                                                                                                     | 1       | 2½      | 2            |
| Generators, batteries, inverters, motors, transformers, and other electrical components constructed of high-deformability materials                                                                                                                   | 1       | 2½      | 2            |
| Motor control centers, panel boards, switch gear, instrumentation cabinets, and other components constructed of sheet metal framing                                                                                                                   | 2½      | 6       | 2            |
| Communication equipment, computers, instrumentation, and controls                                                                                                                                                                                     | 1       | 2½      | 2            |
| Roof-mounted stacks, cooling and electrical towers laterally braced below their center of mass                                                                                                                                                        | 2½      | 3       | 2            |
| Roof-mounted stacks, cooling and electrical towers laterally braced above their center of mass                                                                                                                                                        | 1       | 2½      | 2            |
| Lighting fixtures                                                                                                                                                                                                                                     | 1       | 1½      | 2            |
| Other mechanical or electrical components                                                                                                                                                                                                             | 1       | 1½      | 2            |
| <b>VIBRATION-ISOLATED COMPONENTS AND SYSTEMS<sup>b</sup></b>                                                                                                                                                                                          |         |         |              |
| Components and systems isolated using neoprene elements and neoprene isolated floors with built-in or separate elastomeric snubbing devices or resilient perimeter stops                                                                              | 2½      | 2½      | 2            |
| Spring-isolated components and systems and vibration-isolated floors closely restrained using built-in or separate elastomeric snubbing devices or resilient perimeter stops                                                                          | 2½      | 2       | 2            |
| Internally isolated components and systems                                                                                                                                                                                                            | 2½      | 2       | 2            |
| Suspended vibration-isolated equipment including in-line duct devices and suspended internally isolated components                                                                                                                                    | 2½      | 2½      | 2            |
| <b>DISTRIBUTION SYSTEMS</b>                                                                                                                                                                                                                           |         |         |              |
| Piping in accordance with ASME B31 (2001, 2002, 2008, and 2010), including in-line components with joints made by welding or brazing                                                                                                                  | 2½      | 12      | 2            |
| Piping in accordance with ASME B31, including in-line components, constructed of high- or limited-deformability materials, with joints made by threading, bonding, compression couplings, or grooved couplings                                        | 2½      | 6       | 2            |
| Piping and tubing not in accordance with ASME B31, including in-line components, constructed of high-deformability materials, with joints made by welding or brazing                                                                                  | 2½      | 9       | 2            |
| Piping and tubing not in accordance with ASME B31, including in-line components, constructed of high- or limited-deformability materials, with joints made by threading, bonding, compression couplings, or grooved couplings                         | 2½      | 4½      | 2            |
| Piping and tubing constructed of low-deformability materials, such as cast iron, glass, and nonductile plastics                                                                                                                                       | 2½      | 3       | 2            |
| Ductwork, including in-line components, constructed of high-deformability materials, with joints made by welding or brazing                                                                                                                           | 2½      | 9       | 2            |
| Ductwork, including in-line components, constructed of high- or limited-deformability materials with joints made by means other than welding or brazing                                                                                               | 2½      | 6       | 2            |
| Ductwork, including in-line components, constructed of low-deformability materials, such as cast iron, glass, and nonductile plastics                                                                                                                 | 2½      | 3       | 2            |
| Electrical conduit and cable trays                                                                                                                                                                                                                    | 2½      | 6       | 2            |
| Bus ducts                                                                                                                                                                                                                                             | 1       | 2½      | 2            |
| Plumbing                                                                                                                                                                                                                                              | 1       | 2½      | 2            |
| Pneumatic tube transport systems                                                                                                                                                                                                                      | 2½      | 6       | 2            |

<sup>a</sup>A lower value for  $a_p$  is permitted where justified by detailed dynamic analyses. The value for  $a_p$  shall not be less than 1. The value of  $a_p$  equal to 1 is for rigid components and rigidly attached components. The value of  $a_p$  equal to 2½ is for flexible components and flexibly attached components.

<sup>b</sup>Components mounted on vibration isolators shall have a bumper restraint or snubber in each horizontal direction. The design force shall be taken as  $2F_p$  if the nominal clearance (air gap) between the equipment support frame and restraint is greater than 0.25 in. (6 mm). If the nominal clearance specified on the construction documents is not greater than 0.25 in. (6 mm), the design force is permitted to be taken as  $F_p$ .

<sup>c</sup>Overstrength as required for anchorage to concrete and masonry. See Section 12.4.3 for seismic load effects including overstrength.

3. The connection to the structure shall allow a 360-deg range of motion in the horizontal plane.

Where design of mechanical and electrical components for seismic effects is required, consideration shall be given to the dynamic effects of the components, their contents, and where appropriate, their supports and attachments. In such cases, the interaction between the components and the supporting structures, including other mechanical and electrical components, shall also be considered.

**13.6.2 Mechanical Components.** HVACR ductwork shall meet the requirements of Section 13.6.6. Piping systems shall meet the requirements of Section 13.6.7. Boilers and vessels shall meet the requirements of Section 13.6.10. Elevators shall meet the requirements of Section 13.6.11. All other mechanical components shall meet the requirements of Section 13.6.13. Mechanical components with  $I_p$  greater than 1.0 shall be designed for the seismic forces and relative displacements defined in Sections 13.3.1 and 13.3.2 and shall satisfy the following additional requirements:

1. Provision shall be made to eliminate seismic impact for components vulnerable to impact, for components constructed of nonductile materials, and in cases where material ductility will be reduced because of service conditions (e.g., low-temperature applications).
2. The possibility of loads imposed on components by attached utility or service lines, caused by differential movement of support points on separate structures, shall be evaluated.
3. Where piping or HVACR ductwork components are attached to structures that could displace relative to one another and for isolated structures where such components cross the isolation interface, the components shall be designed to accommodate the seismic relative displacements defined in Section 13.3.2.

**13.6.2.1 HVACR Equipment.** HVACR equipment that has been qualified in accordance with the requirements of Chapters 1 through 10 of ANSI/AHRI Standard 1270 (I-P) or ANSI/AHRI Standard 1271 (SI) shall be deemed to meet the seismic qualification requirements of Section 13.2.2, provided all of the following requirements are met:

- a. Active and/or energized components shall be seismically certified exclusively through shake table testing or experience data; and
- b. Seismic demand considered in the certification of non-active components through analysis shall be based on  $R_p/I_p$  equal to 1.0; and
- c. Capacity of non-active components used in seismic certification by analysis shall be based on the provisions of ASCE 7; and
- d. Rugged components shall conform to the definition in Chapter 11.

**13.6.3 Electrical Components.** Conduit, cable tray and raceways shall meet the requirements of Section 13.6.5. Utility and service lines shall meet the requirements of Section 13.6.9. Other electrical components shall meet the requirements of Section 13.6.13. All electrical components with  $I_p$  greater than 1.0 shall be designed for the seismic forces and relative displacements defined in Sections 13.3.1 and 13.3.2 and shall satisfy the following additional requirements:

1. Provision shall be made to eliminate seismic impact between components.
2. Loads imposed on the components by attached utility or service lines that are attached to separate structures shall be evaluated.
3. Batteries on racks shall have wraparound restraints to ensure that the batteries do not fall from the racks. Spacers shall be used between restraints and cells to prevent damage to cases. Racks shall be evaluated for sufficient lateral load capacity.
4. Internal coils of dry type transformers shall be positively attached to their supporting substructure within the transformer enclosure.
5. Electrical control panels, computer equipment, and other items with slide-out components shall have a latching mechanism to hold the components in place.
6. Electrical cabinet design shall comply with the applicable National Electrical Manufacturers Association (NEMA) standards. Cutouts in the lower shear panel that have not been made by the manufacturer and reduce significantly the strength of the cabinet shall be specifically evaluated.
7. The attachments for additional external items weighing more than 100 lb (445 N) shall be specifically evaluated if not provided by the manufacturer.
8. Where conduit, cable trays, or similar electrical distribution components are attached to structures that could displace relative to one another and for isolated structures where such components cross the isolation interface, the components shall be designed to accommodate the seismic relative displacements defined in Section 13.3.2.

**13.6.4 Component Supports.** Mechanical and electrical component supports (including those with  $I_p = 1.0$ ) and the means by which they are attached to the component shall be designed for the forces and displacements determined in Sections 13.3.1 and 13.3.2. Such supports include structural members, braces, frames, skirts, legs, saddles, pedestals, cables, guys, stays, snubbers, tethers, and elements forged or cast as a part of the mechanical or electrical component.

**13.6.4.1 Design Basis.** If standard supports, for example, ASME B31, NFPA 13, or MSS SP-58, or proprietary supports are used, they shall be designed by either load rating (i.e., testing) or for the calculated seismic forces. In addition, the stiffness of the support, where appropriate, shall be designed such that the seismic load path for the component performs its intended function.

**13.6.4.2 Design for Relative Displacement.** Component supports shall be designed to accommodate the seismic relative displacements between points of support determined in accordance with Section 13.3.2.

**13.6.4.3 Support Attachment to Component.** The means by which supports are attached to the component, except where integral (i.e., cast or forged), shall be designed to accommodate both the forces and displacements determined in accordance with Sections 13.3.1 and 13.3.2. If the value of  $I_p = 1.5$  for the component, the local region of the support attachment point to the component shall be evaluated for the effect of the load transfer on the component wall.

**13.6.4.4 Material Detailing Requirements.** The materials comprising supports and the means of attachment to the component shall be constructed of materials suitable for the application, including the effects of service conditions, for

example, low-temperature applications. Materials shall be in conformance with a nationally recognized standard.

**13.6.4.5 Additional Requirements.** The following additional requirements shall apply to mechanical and electrical component supports:

1. Seismic supports shall be constructed so that support engagement is maintained.
2. Reinforcement (e.g., stiffeners or Belleville washers) shall be provided at bolted connections through sheet metal equipment housings as required to transfer the equipment seismic loads specified in this section from the equipment to the structure. Where equipment has been certified per Section 13.2.2, 13.2.5, or 13.2.6, anchor bolts or other fasteners and associated hardware as included in the certification shall be installed in conformance with the manufacturer's instructions. For those cases where no certification exists or where instructions for such reinforcement are not provided, reinforcement methods shall be as specified by a registered design professional or as approved by the Authority Having Jurisdiction.
3. Where weak-axis bending of cold-formed steel supports is relied on for the seismic load path, such supports shall be specifically evaluated.
4. Components mounted on vibration isolators shall have a bumper restraint or snubber in each horizontal direction, and vertical restraints shall be provided where required to resist overturning. Isolator housings and restraints shall be constructed of ductile materials. (See additional design force requirements in footnote *b* to Table 13.6-1.) A viscoelastic pad or similar material of appropriate thickness shall be used between the bumper and components to limit the impact load.

**13.6.5 Distribution Systems: Conduit, Cable Tray, and Raceways.** Cable trays and raceways shall be designed for seismic forces and seismic relative displacements as required in Section 13.3. Conduit greater than 2.5 in. (64 mm) trade size and attached to panels, cabinets, or other equipment subject to seismic relative displacement,  $D_{pl}$ , shall be provided with flexible connections or designed for seismic forces and seismic relative displacements as required in Section 13.3.

#### EXCEPTIONS:

1. Design for the seismic forces and relative displacements of Section 13.3 shall not be required for raceways with  $I_p = 1.0$  where flexible connections or other assemblies are provided between the cable tray or raceway and associated components to accommodate the relative displacement, where the cable tray or raceway is positively attached to the structure, and where one of the following apply:
  - a. Trapeze assemblies are used with 3/8-in. (10-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the conduit, cable tray, or raceway support point to the connection at the supporting structure to support raceways, and the total weight supported by any single trapeze is 100 lb (445 N) or less, or
  - b. Trapeze assemblies with 1/2-in. (13-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the conduit, cable tray, or raceway support point to the connection at the supporting structure are used to support the cable tray or raceway, and the total weight supported by any single trapeze is 200 lb (890 N) or less, or

- c. Trapeze assemblies with 1/2-in. (13-mm) diameter rod hangers not exceeding 24 in. (610 mm) in length from the conduit, cable tray, or raceway support point to the connection at the supporting structure are used to support the cable tray or raceway, and the total weight supported by any single trapeze is 100 lb (445 N) or less, or
  - d. The conduit, cable tray, or raceway is supported by individual rod hangers 3/8 in. (10 mm) or 1/2 in. (13 mm) in diameter, and each hanger in the raceway run is 12 in. (305 mm) or less in length from the conduit, cable tray, or raceway support point connection to the supporting structure, and the total weight supported by any single rod is 50 lb (220 N) or less.
2. Design for the seismic forces and relative displacements of Section 13.3 shall not be required for conduit, regardless of the value of  $I_p$ , where the conduit is less than 2.5 in. (64 mm) trade size.

Design for the displacements across seismic joints shall be required for conduit, cable trays, and raceways with  $I_p = 1.5$  without consideration of conduit size.

**13.6.6 Distribution Systems: Duct Systems.** HVACR and other duct systems shall be designed for seismic forces and seismic relative displacements as required in Section 13.3.

**EXCEPTIONS:** The following exceptions pertain to ductwork not designed to carry toxic, highly toxic, or flammable gases or not used for smoke control:

1. Design for the seismic forces and relative displacements of Section 13.3 shall not be required for duct systems with  $I_p = 1.0$  where flexible connections or other assemblies are provided to accommodate the relative displacement between the duct system and associated components, the duct system is positively attached to the structure, and where one of the following apply:
  - a. Trapeze assemblies with 3/8-in. (10-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the duct support point to the connection at the supporting structure are used to support duct, and the total weight supported by any single trapeze is less than 10 lb/ft (146 N/m); or
  - b. Trapeze assemblies with 1/2-in. (13-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the duct support point to the connection at the supporting structure are used to support the duct, and the total weight supported by any single trapeze is 200 lb (890 N) or less, or
  - c. Trapeze assemblies with 1/2-in. (13-mm) diameter rod hangers not exceeding 24 in. (610 mm) in length from the duct support point to the connection at the supporting structure are used to support the duct, and the total weight supported by any single trapeze is 100 lb (445 N) or less, or
  - d. The duct is supported by individual rod hangers 3/8 in. (10 mm) or 1/2 in. (13 mm) in diameter, and each hanger in the duct run is 12 in. (305 mm) or less in length from the duct support point to the connection at the supporting structure, and the total weight supported by any single rod is 50 lb (220 N) or less.
2. Design for the seismic forces and relative displacements of Section 13.3 shall not be required where provisions are made to avoid impact with other ducts or mechanical components or to protect the ducts in the event of such impact, the distribution system is positively attached to the structure;

and HVACR ducts have a cross-sectional area of less than 6 ft<sup>2</sup> (0.557 m<sup>2</sup>) and weigh 20 lb/ft (292 N/m) or less.

Components that are installed in line with the duct system and have an operating weight greater than 75 lb (334 N), such as fans, terminal units, heat exchangers, and humidifiers, shall be supported and laterally braced independent of the duct system, and such braces shall meet the force requirements of Section 13.3.1. Components that are installed in line with the duct system, have an operating weight of 75 lb (334 N) or less, such as small terminal units, dampers, louvers, and diffusers, and are otherwise not independently braced shall be positively attached with mechanical fasteners to the rigid duct on both sides. Piping and conduit attached to in-line equipment shall be provided with adequate flexibility to accommodate the seismic relative displacements of Section 13.3.2.

**13.6.7 Distribution Systems: Piping and Tubing Systems.** Unless otherwise noted in this section, piping and tubing systems shall be designed for the seismic forces and seismic relative displacements of Section 13.3. ASME pressure piping systems shall satisfy the requirements of Section 13.6.7.1. Fire protection sprinkler piping shall satisfy the requirements of Section 13.6.7.2. Elevator system piping shall satisfy the requirements of Section 13.6.11.

Where other applicable material standards or recognized design bases are not used, piping design including consideration of service loads shall be based on the following allowable stresses:

- for piping constructed with ductile materials (e.g., steel, aluminum, or copper), 90% of the minimum specified yield strength;
- for threaded connections in piping constructed with ductile materials, 70% of the minimum specified yield strength;
- for piping constructed with nonductile materials (e.g., cast iron or ceramics), 10% of the material minimum specified tensile strength; and
- for threaded connections in piping constructed with nonductile materials, 8% of the material minimum specified tensile strength.

Piping not detailed to accommodate the seismic relative displacements at connections to other components shall be provided with connections that have sufficient flexibility to avoid failure of the connection between the components.

Suspended components that are installed in line and rigidly connected to and supported by the piping system, such as valves, strainers, traps, pumps, air separators, and tanks, are permitted to be considered part of the piping system for the purposes of determining the need for and sizing of lateral bracing. Where components are braced independently because of their weight but the associated piping is not braced, flexibility shall be provided as required to accommodate relative movement between the components.

**13.6.7.1 ASME Pressure Piping Systems.** Pressure piping systems, including their supports, designed and constructed in accordance with ASME B31 shall be deemed to meet the force, displacement, and other requirements of this section. In lieu of specific force and displacement requirements provided in ASME B31, the force and displacement requirements of Section 13.3 shall be used. Materials meeting the toughness requirements of ASME B31 shall be considered high-deformability materials.

**13.6.7.2 Fire Protection Sprinkler Piping Systems.** Fire protection sprinkler piping, pipe hangers, and bracing

designed and constructed in accordance with NFPA 13 shall be deemed to meet the force and displacement requirements of this section. Clearances for sprinkler drops and sprigs and other equipment shall conform to 13.2.3.1. The exceptions of Section 13.6.7.3 shall not apply.

**13.6.7.3 Exceptions.** Design for the seismic forces of Section 13.3 shall not be required for piping systems where flexible connections, expansion loops, or other assemblies are provided to accommodate the relative displacement between component and piping, where the piping system is positively attached to the structure, and where one of the following apply:

- Trapeze assemblies are used to support piping whereby no single pipe exceeds the limits set forth in 5a, 5b, or 5c below and the total weight of the piping supported by the trapeze assemblies is less than 10 lb/ft (146 N/m).
- Trapeze assemblies are supported by 3/8-in. (10-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the pipe support point to the connection at the supporting structure, do not support piping with  $I_p$  greater than 1.0, and no single pipe exceeds the limits set forth in items 5a, 5b, or 5c below and the total weight supported by any single trapeze is 100 lb (445 N) or less, or
- Trapeze assemblies are supported by 1/2-in. (13-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the pipe support point to the connection at the supporting structure, do not support piping with  $I_p$  greater than 1.0, and no single pipe exceeds the diameter limits set forth in items 5a, 5b, or 5c below and the total weight supported by any single trapeze is 200 lb (890 N) or less, or
- Trapeze assemblies are supported by 1/2-in. (13-mm) diameter rod hangers not exceeding 24 in. (610 mm) in length from the pipe support point to the connection at the supporting structure, do not support piping with  $I_p$  greater than 1.0, and no single pipe exceeds the diameter limits set forth in items 5a, 5b, or 5c below and the total weight supported by any single trapeze is 100 lb (445 N) or less, or
- Piping that has an  $R_p$  in Table 13.6-1 of 4.5 or greater is either supported by rod hangers and provisions are made to avoid impact with other structural or nonstructural components or to protect the piping in the event of such impact, or pipes with  $I_p = 1.0$  are supported by individual rod hangers 3/8 in. (10 mm) or 1/2 in. (13 mm) in diameter; where each hanger in the pipe run is 12 in. (305 mm) or less in length from the pipe support point to the connection at the supporting structure; and the total weight supported by any single hanger is 50 lb (220 N) or less. In addition, the following limitations on the size of piping shall be observed:
  - In structures assigned to Seismic Design Category C where  $I_p$  is greater than 1.0, the nominal pipe size shall be 2 in. (50 mm) or less.
  - In structures assigned to Seismic Design Categories D, E, or F where  $I_p$  is greater than 1.0, the nominal pipe size shall be 1 in. (25 mm) or less.
  - In structures assigned to Seismic Design Categories D, E, or F where  $I_p = 1.0$ , the nominal pipe size shall be 3 in. (80 mm) or less.
- Pneumatic tube systems supported with trapeze assemblies using 3/8-in. (10-mm) diameter rod hangers not exceeding 12 in. (305 mm) in length from the tube support point to the connection at the supporting structure and the total weight supported by any single trapeze is 100 lb (445 N) or less.
- Pneumatic tube systems supported by individual rod hangers 3/8 in. (10 mm) or 1/2 in. (13 mm) in diameter, and



each hanger in the run is 12 in. (305 mm) or less in length from the tube support point to the connection at the supporting structure, and the total weight supported by any single rod is 50 lb (220 N) or less.

**13.6.8 Distribution Systems: Trapezes with a Combination of Systems.** Trapezes that support a combination of distribution systems (electrical conduit, raceway, duct, piping, etc.) shall be designed using the most restrictive requirements for the supported distribution systems from Sections 13.6.5 through 13.6.8 for the aggregate weight of the supported system. If any distribution system on the trapeze is not exempted, the trapeze shall be braced.

**13.6.9 Utility and Service Lines.** At the interface of adjacent structures or portions of the same structure that may move independently, utility lines shall be provided with adequate flexibility to accommodate the anticipated differential movement between the portions that move independently. Differential displacement calculations shall be determined in accordance with Section 13.3.2.

The possible interruption of utility service shall be considered in relation to designated seismic systems in Risk Category IV as defined in Table 1.5-1. Specific attention shall be given to the vulnerability of underground utilities and utility interfaces between the structure and the ground where Site Class E or F soil is present, and where the seismic coefficient  $S_{DS}$  at the underground utility or at the base of the structure is equal to or greater than 0.33.

**13.6.10 Boilers and Pressure Vessels.** Boilers or pressure vessels designed and constructed in accordance with ASME BPVC shall be deemed to meet the force, displacement, and other requirements of this section. In lieu of the specific force and displacement requirements provided in ASME BPVC, the force and displacement requirements of Sections 13.3.1 and 13.3.2 shall be used. Materials that meet the toughness requirements of ASME BPVC shall be considered high-deformability materials. Other boilers and pressure vessels designated as having an  $I_p = 1.5$ , but not designed and constructed in accordance with the requirements of ASME BPVC, shall comply with the requirements of Section 13.6.13.

**13.6.11 Elevator and Escalator Design Requirements.** Elevators and escalators designed in accordance with the seismic requirements of ASME A17.1 shall be deemed to meet the seismic force requirements of this section, except as modified in the following text. The exceptions of Section 13.6.7.3 shall not apply to elevator piping.

**13.6.11.1 Escalators, Elevators, and Hoistway Structural Systems.** Escalators, elevators, and hoistway structural systems shall be designed to meet the force and displacement requirements of Sections 13.3.1 and 13.3.2.

**13.6.11.2 Elevator Equipment and Controller Supports and Attachments.** Elevator equipment and controller supports and attachments shall be designed to meet the force and displacement requirements of Sections 13.3.1 and 13.3.2.

**13.6.11.3 Seismic Controls for Elevators.** Elevators operating with a speed of 150 ft/min (46 m/min) or greater shall be provided with seismic switches. Seismic switches shall provide an electric signal indicating that structural motions are of such a magnitude that the operation of the elevators may be impaired. Seismic switches in accordance with Section 8.4.10.1.2 of ASME A17.1 shall be deemed to meet the requirements of this section.

**EXCEPTION:** In cases where seismic switches cannot be located near a column in accordance with ASME A17.1, they shall have two horizontal axes of sensitivity and have a trigger level set to 20% of the acceleration of gravity where located at or near the base of the structure and 50% of the acceleration of gravity in all other locations.

Upon activation of the seismic switch, elevator operations shall conform to requirements of ASME A17.1, except as noted in the following text.

In facilities where the loss of the use of an elevator is a life-safety issue, the elevator shall only be used after the seismic switch has triggered, provided that

1. The elevator shall operate no faster than the service speed, and
2. Before the elevator is occupied, it is operated from top to bottom and back to top to verify that it is operable.

**13.6.11.4 Retainer Plates.** Retainer plates are required at the top and bottom of the car and counterweight.

**13.6.12 Rooftop Solar Panels.** Rooftop solar panels and their attachments shall be designed for the forces and displacements determined in Section 13.3.

**EXCEPTION:** Ballasted solar panels without positive direct attachment to the roof structure are permitted on Risk Category I, II, and III structures six stories or fewer in height and having a maximum roof slope equal to or less than 1 in 20, provided that they comply with the following:

1. The height of the center of mass of any panel above the roof surface is less than half the least spacing in plan of the panel supports, but in no case greater than 3 ft (0.9 m).
2. Each panel is designed to accommodate without impact, instability, or loss of support a seismic displacement,  $\delta_{mpv}$ , of the panel relative to any roof edge or offset and any other curb or obstruction to sliding on the roof surface where  $\delta_{mpv}$  is determined in accordance with Eq. (13.6-1), but is not taken as less than 2 ft (1.2 m):

$$\delta_{mpv} = 5I_e(S_{DS} - 0.4)^2[\text{ft(m)}] \quad (13.6-1)$$

The minimum separation between adjacent unattached panels shall be taken as  $0.5\delta_{mpv}$ . Signage or roof markings (e.g., yellow stripes) shall be provided delineating the area around the panel that must be kept free of obstructions. Alternatively,  $\delta_{mpv}$  may be determined by shake table testing or nonlinear response history analysis, whereby the value of  $\delta_{mpv}$  shall not be taken as less than 80% of the value given by Eq. (13.6-1) unless independent peer review is conducted in accordance with Section 1.3.1.3.4.

3. Each panel is interconnected to resist a horizontal force of  $0.2S_{DS}W_{pi}$ , across any section cut by a vertical plane, where  $W_{pi}$  is the weight of the smaller of the two portions.
4. Panel framing and supports are designed for a seismic force path from the center of mass of each component to locations of friction resistance equal to the lesser of  $F_p$  from Section 13.3.1 and  $0.6W_p$ , where  $W_p$  is the weight of each component.
5. All electrical cables leading from a panel to another panel or to another roof object are designed to accommodate, without rupture or distress, differential movements between cable connection points of  $1.0\delta_{mpv}$ , with consideration given to torsional movement of the panel and its possible impingement on the electrical cables.

6. All edges and offsets of roof surfaces on which panels are placed are bounded by a curb or parapet not less than 12 in. (0.3 m) in height and designed to resist a concentrated load applied at the probable points of impact between the curb or parapet and the panel of not less than  $0.2S_{DS}$  times the weight of the panel. Alternatively, a panel may be placed so that all parts of the panel are a minimum of  $2.0\delta_{mpv}$ , but not less than 4 ft (1.22 m), from any roof edge or offset.
7. Where justified by testing and analysis, the maximum roof slope for structures assigned to SDC C and D shall be permitted to be 1 in 12 provided that independent peer review is conducted in accordance with Section 1.3.1.3.4.

### 13.6.13 Other Mechanical and Electrical Components.

Mechanical and electrical components, including conveyor systems, not designed and constructed in accordance with the reference documents in Chapter 23 shall meet the following:

1. Components and their supports and attachments shall comply with the requirements of Sections 13.4, 13.6.2, 13.6.3, and 13.6.4.
2. For mechanical components with hazardous substances and assigned a component Importance Factor,  $I_p$ , of 1.5 in accordance with Section 13.1.3 and for boilers and

pressure vessels not designed in accordance with ASME BPVC, the design strength for seismic loads in combination with other service loads and appropriate environmental effects shall be based on the following material properties:

- a. for mechanical components constructed with ductile materials (e.g., steel, aluminum, or copper), 90% of the minimum specified yield strength;
- b. for threaded connections in components constructed with ductile materials, 70% of the minimum specified yield strength;
- c. for mechanical components constructed with nonductile materials (e.g., plastic, cast iron, or ceramics), 10% of the material minimum specified tensile strength; and
- d. for threaded connections in components constructed with nonductile materials, 8% of the material minimum specified tensile strength.

## 13.7 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.